

RESEARCH ARTICLE

Sentiment-Aware Portfolio Optimization: CVaR-Based Diversification With Deep Reinforcement Learning

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ABSTRACT This paper presents an innovative portfolio optimization framework that synergistically integrates CVaR-based diversification with Deep Reinforcement Learning (DRL) and sentiment analysis from a Large Language Model (LLM). Traditional optimization methods, such as Mean-Variance Optimization (MVO) and Most Diversified Portfolio (MDP) strategies, often fall short in capturing tail risks and adapting to dynamic market conditions. In contrast, our approach employs the Deep Deterministic Policy Gradient (DDPG) DRL algorithm to dynamically adjust asset allocations taking transaction costs into account while explicitly managing downside risk via Conditional Value-at-Risk (CVaR). To further enhance the decision-making process, sentiment scores derived from the FinBERT LLM are incorporated into the state space, enabling the model to capture qualitative market signals and improve risk-adjusted performance. Empirical evaluations reveal that our portfolio, the DRL Sentiment-Aware CVaR Ratio Portfolio, outperforms benchmark strategies, including Equally Weighted, Minimum CVaR, and traditional MDP portfolios, by achieving higher annualized returns, improved Sharpe and Omega ratios, and elevated Jensen's alpha. These promising results underscore the potential of combining robust risk management measures for portfolio diversification with advanced machine learning techniques and sentiment analysis to navigate the complexities of contemporary financial markets. Future research directions include extending the framework to multi-asset classes, integrating additional alternative data sources, and exploring alternative DRL algorithms to further enhance performance.

INDEX TERMS Portfolio optimization, conditional value-at-risk (CVaR), most diversified portfolio (MDP), CVaR diversification ratio, deep reinforcement learning (DRL), deep deterministic policy gradient (DDPG), sentiment analysis, FinBERT.

I. INTRODUCTION

Portfolio optimization has long been a central area of interest in finance and a foundational concept in financial economics. Recent research in this field has contributed to the formulation of advanced investment strategies that systematically balance portfolio risk and return. The seminal work of Markowitz (1952) laid the foundation through mean-variance optimization (MVO) which provided a systematic framework for balancing risk and return [1]. Markowitz argues that investors should consider expected returns as desirable and variance or risk as undesirable, hence advocating the

concept of diversification to reduce portfolio risk. Various researchers have used diversification in portfolio construction to build risk-efficient portfolios. Noticeable work in the use of diversification as a drive of portfolio construction includes work by Lehmann and Modest [2], Choueifaty and Coignard [3], Jaeger et al. [4] as well as Choueifaty et al. [5]. These studies have demonstrated the superior results that diversification can achieve in constructing risk-efficient and return-enhancing portfolios.

In advancing the literature on diversification, this study builds on the seminal work of Choueifaty and Coignard [3], who introduced a novel approach to portfolio construction by defining the diversification ratio as the ratio of the weighted average volatility of individual assets to the overall portfolio

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volatility. Their framework aimed to optimize this metric to construct portfolios that maximize diversification and minimize concentration risk. While their approach has proven effective in promoting risk-efficiency, it relies on standard deviation as the risk measure, which may not fully capture the tail risks inherent in financial markets. This study introduces an innovative modification to their framework by incorporating Conditional Value-at-Risk (CVaR), a more nuanced risk metric that focuses on the expected losses in the tails of the return distribution. By replacing standard deviation with CVaR in the diversification ratio, thereby creating a CVaR diversification ratio, the proposed approach seeks to enhance portfolio resilience against extreme market events while maintaining the core objective of maximizing diversification. This integration of CVaR into the diversification framework not only refines the risk measure but also aligns the optimization process more closely with the objectives of risk-averse investors in dynamic and volatile markets.

To further enhance the robustness of our portfolio optimization approach, we integrate machine learning techniques into the rebalancing process, specifically employing deep reinforcement learning (DRL) to address the dynamic nature of financial markets. Deep reinforcement learning has emerged as a powerful framework for dynamic portfolio optimization, enabling agents to learn optimal allocation strategies through iterative interaction with market environments [6], [7]. In our approach, a DRL agent based on the Deep Deterministic Policy Gradient (DDPG) algorithm, interacts with the financial market by observing historical stock prices and sentiment scores, to make asset allocation decisions with the aim of maximizing the CVaR diversification ratio, returns and the utilization of sentiment while penalizing high transaction costs. Sentiment scores are generated using FinBERT [8], a domain-specific large language model (LLM) fine-tuned for financial contexts and incorporating these sentiment scores, our rebalancing process accounts for investor sentiment and market mood, providing a more comprehensive and forward-looking approach to portfolio management. This integration of CVaR-based diversification with DRL-driven adaptive learning enables our framework to efficiently maximize portfolio diversification while addressing tails risks and capturing the complexities and uncertainties of real-world financial markets. Thus, the innovative aspects of our framework can be summarized as follows:

- Improved portfolio diversification and tail risk management through CVaR diversification ratio: By integrating CVaR into the diversification ratio, our model directly addresses the downside risk that traditional variance-based metrics may overlook. This leads to portfolios that are better diversified and more resilient to extreme market movements.
- Dynamic adaptation through DRL: Unlike static optimization methods, our DRL-based approach continually refines its portfolio allocation strategy in real time while penalizing high transaction costs to due frequent

rebalancing. This dynamic adaptation allows for more efficient navigation of volatile markets thereby enhancing the robustness of our investment strategy.

- Enhanced decision-making with sentiment analysis: The incorporation of FinBERT-derived sentiment scores into the state space enriches the information available to the DRL agent. This enables the model to capture the nuanced effects of market sentiment on asset prices, thereby improving risk-adjusted returns.

Empirical evaluations demonstrate that our proposed portfolio, the DRL Sentiment-Aware CVaR Ratio Portfolio, consistently outperforms benchmark strategies, including the simple CVaR Ratio Portfolio, Equally Weighted Portfolio, traditional Most Diversified Portfolio (MDP) and Minimum CVaR portfolios. Notably, the DRL Sentiment-Aware CVaR Ratio Portfolio achieves superior net of trading costs performance metrics, including higher annualized returns (34.57%), Sharpe ratio (1.05), Omega ratio (1.18), and Jensen's alpha (0.38%), when compared to benchmark portfolios. The outstanding results achieved in this study highlight the power of integrating traditional portfolio construction methods, such as the CVaR diversification ratio, with cutting-edge machine learning approaches, including Deep Reinforcement Learning (DRL) and LLM-based sentiment analysis. This fusion of traditional and advanced techniques has proven to be a highly effective strategy for optimizing portfolio performance.

The remainder of the paper is organized as follows. Section II reviews the relevant literature and situates our contribution within the broader context of portfolio optimization research. Section III details the theoretical underpinnings of our CVaR Ratio Portfolio and the technical implementation of the DRL framework, including the integration of sentiment analysis. Section IV presents a comprehensive performance evaluation of the proposed model against established benchmarks such as the simple CVaR Ratio Portfolio, the Equally Weighted Portfolio, and the Minimum CVaR portfolio including the impact of transaction costs, performance sensitivity analysis and robustness checks. Finally, Section V concludes with a discussion of the implications of our findings and directions for future research. In an era marked by unprecedented market complexity and rapid technological advancements, our approach represents a significant step forward in portfolio optimization. By effectively combining deep reinforcement learning, a sophisticated risk measure, and state-of-the-art sentiment analysis, the proposed framework not only enhances predictive accuracy and adaptability but also provides investors with a robust tool for managing risk and achieving superior long-term performance.

II. LITERATURE REVIEW

Portfolio optimization has been a pivotal area in financial economics, with Markowitz's [1] introduction of mean-variance optimization (MVO) being regarded as one

of the cornerstones of portfolio optimization. MVO has been widely adopted in practice, but its underlying assumptions including normal distribution of returns and constant covariance matrices often fail to capture the complexities of real-world markets. Furthermore, the quadratic utility function used in MVO can lead to concentrated positions by prioritizing assets with high expected returns, while estimation errors in expected returns can also result in sub-optimal portfolio allocations. These limitations of MVO can increase concentration risk, prompting market practitioners and researchers to explore alternative optimization strategies. One such approach is the Most Diversified Portfolio (MDP) framework, introduced by Chouifaty and Coignard, which seeks to mitigate concentration risk by maximizing the diversification ratio [3]. Here, diversification is seen as an inherent risk mitigation strategy, somewhat akin to MVO's variance reduction but with a focus on spreading risk across more dimensions than just variance.

While Chouifaty and Coignard [3] showed the Most Diversified Portfolio outperforms other quantitative portfolios using US equity data, comparative studies on the Most Diversified Portfolio show a mixed picture. A study by Theron and Van Vuuren [9] found that the Most Diversified Portfolio under-performed other quantitative portfolios as assets could not be diversified effectively in an emerging market setting. In a study comparing the performance of risk-based portfolios during the 2007/2008 financial crises, Muteba and Mantshimuli showed that the Most Diversified Portfolio was one of the top-performing portfolios during and post the crisis as correlations between assets declined [10]. Cesarone et al. showed that their bi-objective model for portfolio selection, which maximises both a diversification measure and the portfolio expected return, outperforms strategies based only on the diversification measure such as the Most Diversified Portfolio and other classical risk-return approaches [11]. While there is evidence that the Most Diversified Portfolio performs well in certain circumstances, it does not inherently address tail risk as it uses variance or standard deviation as a risk measure.

This gap is addressed by Conditional Value-at-Risk (CVaR), introduced by Rockafellar and Uryasev, which focuses on the expected loss beyond a given confidence level, thus directly tackling tail risk [12]. CVaR provides a more nuanced risk assessment compared to variance, particularly in non-Gaussian markets where extreme events are not well captured by variance alone. The work by Krokmal, Palmquist, and Uryasev illustrates how CVaR can lead to portfolios that are not only diversified but also resilient to market shocks, enhancing the ideas from both MVO and MDP [13]. From a computational standpoint, Monte Carlo methods are widely employed for estimating, analyzing sensitivity, and optimizing CVaR, offering a robust framework for financial risk management, especially when loss distributions lack closed-form expressions [44]. Further technical advancements include robust optimization techniques for mean-CVaR portfolio selection, addressing esti-

mation risk in mean returns [43], and novel sampling-based estimators for CVaR gradients, allowing for optimization in more complex domains, including reinforcement learning applications [38]. Alexander and Baptista further highlight CVaR's advantages over variance in portfolio optimization by demonstrating superior out-of-sample performance under conditions of return estimation uncertainty [14]. Portfolio optimization using CVaR has been considered under different investment conditions, including optimizing portfolio risk in the presence of non-normal returns [15], managing risk in emerging markets [16], and construction of portfolios in the cryptocurrency market [17].

Building on the concepts of Conditional Value-at-Risk (CVaR) and the Most Diversified Portfolio (MDP), a recent approach has been proposed that combines the benefits of diversification and tail-risk management. Specifically, Cesarone et al. introduced the concept of the CVaR maximum diversification ratio, which optimizes the ratio of the portfolio's expected return to its Conditional Value-at-Risk (CVaR) while maximizing diversification [11]. By incorporating CVaR into the diversification ratio, this approach offers a more comprehensive framework for portfolio optimization, one that balances the trade-off between expected return, tail risk, and diversification. The study by Cesarone et al. demonstrated enhanced out-of-sample performance, with their method surpassing strategies that rely solely on diversification or the traditional risk-return approach [11]. While diversification-based portfolio optimization techniques have shown promising results, the fusion of these techniques with cutting-edge computational paradigms, such as machine learning (ML), reinforcement learning (RL), and natural language processing (NLP), opens up highly promising avenues for advancing research in portfolio optimization.

These emerging technologies have demonstrated significant potential in enhancing traditional portfolio optimization approaches, including the Markowitz mean-variance optimization [18], Black-Litterman framework [19] and multi-factor portfolio optimization [20], by addressing their inherent limitations and improving their predictive accuracy and adaptability. Machine learning techniques, for instance, have been leveraged to capture complex, non-linear relationships within financial data that traditional models often overlook. Algorithms such as neural networks and support vector machines have been applied to improve risk-adjusted returns by identifying hidden patterns in asset price movements and macroeconomic indicator [21]. Reinforcement learning, on the other hand, has shown promise in dynamically optimizing portfolios by learning optimal strategies through trial and error in simulated environments, thereby adapting to changing market conditions more effectively than static models [6].

Recent advancements highlight DRL's efficacy in managing portfolios, particularly in dynamic and volatile financial markets, by not relying on fixed assumptions and adapting through learning processes [45]. These models can address complex objectives beyond just profit, such as risk and

sustainability, and are well-suited for multi-stage decision-making in financial contexts ([25], [46]). DRL frameworks have also been developed to provide financial-model-free solutions for portfolio management, integrating prediction and allocation steps into a unified autonomous system [47]. Natural language processing has further enriched portfolio optimization by extracting actionable insights from unstructured data sources, such as news articles, earnings reports, and social media sentiment, which can be integrated into portfolio decision-making processes [22]. Specifically, domain-specific models like FinBERT have revolutionized financial sentiment analysis, outperforming generic models and traditional machine learning methods in classifying sentiment from financial texts, including news, reports, and social media ([8], [48], [49]). Empirical studies demonstrate their utility in predicting stock market movements and integrating sentiment factors into asset pricing models for more nuanced market dynamics analysis [40], [50].

These computational paradigms have not only enhanced the performance of traditional approaches but also introduced novel methodologies that account for real-world complexities, such as transaction costs, liquidity constraints, and behavioral biases. For example, hybrid models combining machine learning (ML) with Markowitz optimization have demonstrated superior performance over conventional methods in both in-sample and out-of-sample testing, taking into account transaction costs [23]. It is important to take transaction costs into account in portfolio optimization studies as this breaches the gap between theory and practice, as highlighted by Buccheri et al. [42]. Our study takes into account transaction costs in two ways: first, the reward function penalizes excessive trading costs by minimizing turnovers (similar to Pretorius and van Zyl, 2022 [25]), and secondly, the performance of the DRL CvaR ratio portfolio and its benchmarks are measured net of transactions or trading costs.

Similarly, multi-factor models augmented with NLP-driven sentiment analysis have provided deeper insights into market dynamics, enabling more robust asset allocation strategies by incorporating real-time, unstructured data sources like news, social media, and corporate disclosures [24]. These advancements, particularly in financial sentiment analysis using fine-tuned deep learning models, allow for more accurate classification of sentiment from diverse financial texts, significantly enhancing decision-making and the prediction of market movements [39]. Furthermore, integrating text-derived, time-varying sentiment factors from large-scale financial news and social media data into asset pricing models has proven crucial for explaining variations in daily stock returns and adapting to evolving market volatility [40]. A survey of the use cases of machine learning and artificial intelligence in portfolio optimization indicated a growing use of these techniques in areas such as stock selection, asset allocation, risk management, and portfolio rebalancing, with algorithms such as support

vector machines, neural networks, and decision trees being increasingly applied to optimize portfolio performance [26].

The integration of artificial intelligence and machine learning with traditional portfolio optimization methodologies has shown promising results. However, even greater performance enhancements can be achieved by applying machine learning and artificial intelligence to hybrid models that combine multiple traditional optimization techniques, rather than solely focusing on augmenting individual traditional methods. This study seeks to harness the potential of machine learning to enhance portfolio optimization by integrating Deep Reinforcement Learning (DRL) with two traditional optimization techniques: Most Diversified Portfolio (MDP) and Conditional Value-at-Risk (CVaR) portfolio optimization. By leveraging DRL, this approach aims to dynamically adapt to new market data, including stock sentiment, overcoming the limitations of traditional models which can be overly static. Recent studies have demonstrated that DRL-based portfolio strategies can outperform traditional models by effectively capturing time-dependent patterns in financial data [7]. Furthermore, combining DRL with risk-sensitive objective functions, such as CVaR, allows for improved downside risk management [27]. By leveraging alternative data sources such as sentiment scores, DRL models can further refine their decision-making processes and respond to market conditions in real time [28]. Sentiment analysis has been increasingly utilized in financial markets to extract insights from textual data sources, such as financial news, analyst reports, and social media posts.

Studies have consistently demonstrated that investor sentiment significantly influences asset prices and market volatility, with early research highlighting the role of news sentiment and social media in shaping market dynamics [29], [30]. Traditional approaches to sentiment analysis relied on simple text-based metrics, such as word counts and sentiment lexicons, to gauge market sentiment. However, advancements in machine learning (ML) and natural language processing (NLP) have enabled more sophisticated methods for quantifying sentiment and integrating it into trading strategies [31]. For instance, ML models like support vector machines and recurrent neural networks have been used to classify sentiment from financial news and social media, providing actionable insights for portfolio management. In recent years, the advent of large language models (LLMs) such as GPT-4, BERT, and their derivatives has revolutionized sentiment analysis in finance. These models, pre-trained on vast corpora of text data, excel at capturing nuanced linguistic patterns, and context, which are critical for accurately interpreting investor sentiment.

The integration of large language model (LLM) based sentiment analysis into portfolio optimization has been shown to enhance portfolio performance by capturing market sentiment and making more informed investment decisions with portfolios demonstrating improved risk-adjusted returns and robust performance [32], [33]. Furthermore, adapting

general-purpose LLMs such as Bidirectional Encoder Representations from Transformers (BERT) and Large Language Model Meta AI (Llama), through domain-specific fine-tuning has proven critical for financial applications, as specialized models exhibit superior efficacy in decoding market nuances. Notable examples include FinBERT Prosus AI [8] and FinLlama [34], which outperform their base architectures (BERT and Llama, respectively) in sentiment classification tasks, achieving higher accuracy in financial contexts.

For this study, we employ FinBERT (Prosus AI) to derive sentiment scores, prioritizing its established reliability in financial natural language processing (NLP) tasks, widespread adoption in academic research (See Sidogi et al., Lee et al. and Jun et al., [35], [36], [37]), compatibility with real-time market data streams and superior classification accuracy as demonstrated in our validation tests (Section IV-B). Its architecture, pre-trained on corpora encompassing earnings calls, financial news, and regulatory filings, ensures granular sentiment extraction aligned with investor behavior, a feature corroborated by its recurrent use in recent literature. The integration of FinBERT sentiment analysis, deep reinforcement learning (DRL), and CVaR-based Most Diversified Portfolios (MDPs) offers a robust framework for portfolio optimization. The next section of this study outlines the methodology for implementing this framework which includes the development of the CVaR ratio and the application of DRL, guided by sentiment insights from FinBERT, to achieve optimal portfolio allocation.

III. METHODOLOGY

Our methodology integrates three complementary components to optimize portfolio allocation under tail risk constraints. First, we establish the theoretical foundation through CVaR diversification ratio optimization, addressing the limitations of traditional variance-based approaches. Second, we develop a sentiment analysis pipeline to extract market sentiment signals from financial news, providing behavioral indicators that enhance traditional price-based features. Finally, we implement a Deep Reinforcement Learning framework that can navigate the non-convex optimization landscape of CVaR diversification while incorporating both market and sentiment data for improved decision-making.

A. CVAR DIVERSIFICATION RATIO

Consider n different assets and let R_i be a random variable that represents the return of asset i . A portfolio is created by investing in the n available assets and letting R_p be the portfolio return. We have:

$$R_p = \sum_{i=1}^n \omega_i R_i, \quad (1)$$

where ω_i is the weight for asset i , for $i = 1, \dots, n$. The standard deviation of R_i is given by σ_i while the standard deviation of the portfolio returns, R_p is σ_p . Choueifaty and Coignard [3] define the optimal problem based on a

diversification ratio that is defined by

$$DR_{\text{std}} = \frac{\sum_{i=1}^n \omega_i \sigma_i}{\sigma_p} = \frac{\omega' \sigma}{\omega' \Sigma \omega}, \quad (2)$$

where $\omega = (\omega_1, \dots, \omega_n)'$, σ is a column vector of individual standard deviations σ_i , $i = 1, \dots, n$, and Σ is the variance-covariance matrix of the n returns. The optimal portfolio is selected as the solution to the following optimal problem:

$$\begin{aligned} \omega^* &= \max \ln(DR_{\text{std}}) \\ \text{s.t. } \mathbf{1}'\omega &= 1 \end{aligned}$$

The ratio DR_{std} is the diversification ratio when the main risk measure is the standard deviation. In this paper, we consider extending this problem by using Conditional Value-at-Risk (CVaR) as a risk measure instead of the standard deviation. CVaR is the conditional expected loss above Value-at-Risk (VaR), where VaR is the maximum potential loss over a given period for a defined confidence interval. VaR and CVaR are formulated as follows for a random rate of return of a portfolio (R_p):

$$VaR_\kappa(R_p) = \inf \{R_p \in \mathbb{R}, F(R_p) \geq \kappa\} \quad (3)$$

and

$$CVaR_\kappa(R_p) = \mathbb{E}[R_p \mid R_p < VaR_\kappa(R_p)], \quad (4)$$

where κ is the confidence level with $0 \leq \kappa \leq 1$ and F is the cumulative distribution function of R_p . Using this definition of CVaR, we define the CVaR diversification ratio as:

$$DR_{\text{CVaR}} = \frac{\sum_{i=1}^n \omega_i CVaR_\kappa(R_i)}{CVaR_\kappa(R_p)}. \quad (5)$$

While CVaRs for individual assets, $CVaR_\kappa(R_i)$, can be readily calculated as the average of losses beyond the VaR level using historical or simulated returns, the portfolio CVaR, $CVaR_\kappa(R_p)$, introduces computational complexity. Importantly, the numerator, $\sum_{i=1}^n \omega_i CVaR_\kappa(R_i)$, is a linear combination of convex functions (the individual asset CVaRs) and is therefore convex with respect to the portfolio weights ω . However, direct optimization of Equation (5) may still be non-convex due to the ratio structure and the dependence of the portfolio CVaR on the weights themselves. This non-convexity implies that finding a global optimum with traditional optimization methods can be computationally challenging. To address this, we employ the Rockafellar and Uryasev [12] approximation, which transforms the portfolio CVaR calculation into a convex optimization problem:

$$\tilde{F}_\kappa(\omega, \alpha) = \alpha + \frac{1}{n(1-\kappa)} \sum_{i=1}^n \left[-\omega^T R_i - \alpha \right]^+, \quad (6)$$

where $\tilde{F}_\kappa(\omega, \alpha)$ is the approximation of portfolio CVaR (i.e. $F_\kappa(\omega, \alpha) = CVaR_\kappa(R_p)$) given the portfolio VaR of α and bold symbols representing vectors. This CVaR approximation expression is convex and piecewise linear with respect to α and can be optimized using linear programming. However,

the ratio structure in the diversification ratio still presents challenges for direct optimization. The CVaR diversification ratio optimization faces several numerical stability issues. First, the denominator $CVaR_\kappa(R_p)$ may approach zero or become very small for well-diversified portfolios with low tail risk, leading to numerical instability in the ratio calculation. Second, the approximation error between $\hat{F}_\kappa(\omega, \alpha)$ and the true portfolio CVaR can accumulate, particularly when portfolio weights change significantly during optimization. Third, the piecewise linear nature of the CVaR approximation can introduce discontinuities in the gradient, potentially causing convergence issues in gradient-based optimization methods.

To mitigate these numerical challenges, we implement several stabilization techniques: (1) regularization terms to prevent extreme weight concentrations, (2) bounds on portfolio weights to ensure reasonable diversification, and (3) adaptive scaling of the objective function to maintain numerical precision. Additionally, we employ the Deep Reinforcement Learning framework, which naturally handles non-convex optimization problems through exploration and can adapt to the complex loss landscape of the CVaR diversification ratio. We use the approximation of CVaR in Equation (6) to present the CVaR diversification ratio defined in Equation (5) as follows;

$$DR_{CVaR} = \frac{\sum_{i=1}^n \omega_i (CVaR_\kappa(R_i))}{\alpha + \frac{1}{n(1-\kappa)} \sum_{i=1}^n [-\omega^T R_i - \alpha]^+} \quad (7)$$

Given the CVaR Ratio, DR_{CVaR} , as defined in Equation (7), CVaR Ratio Portfolio can be selected as the solution to the following optimal problems:

$$\begin{aligned} \omega^* &= \max \ln(DR_{CVaR}) \\ \text{s.t. } \mathbf{1}'\omega &= 1. \end{aligned} \quad (8)$$

While this optimization problem is theoretically well-defined, its non-convex nature and numerical complexities motivate our integrated approach combining sentiment analysis and Deep Reinforcement Learning. The sentiment analysis component, detailed in the following subsection, provides additional market signals that can enhance the robustness of portfolio decisions under tail risk constraints.

B. SENTIMENT ANALYSIS PIPELINE

The computation of asset-specific sentiment scores follows a four-stage preparation and analysis pipeline:

a: STAGE 1: DATA COLLECTION

- **Financial news headlines** are sourced from FNSPID Financial News Data Set (publicly available at: <https://huggingface.co/datasets/ashraq/financial-news>).
- **Financial news articles** were obtained from the FNSPID Financial News Data Set (publicly available at: https://github.com/Zdong104/FNSPID_Financial_News_Dataset).
- Data is collected with explicit tracking of publication dates to ensure temporal alignment with market data.

b: STAGE 2: TEXT PREPROCESSING

- **Sentence Segmentation:** Articles are split into sentences using Python NLTK's `sent_tokenize`.
- **Entity Matching:** Case-insensitive regex patterns match S&P 500 constituents using RegEX.
- **Context Preservation:** Only sentences containing matched entities are retained for analysis.
- For pre-annotated datasets (i.e. Ashraq Financial Headlines Data Set), this step is skipped.

c: STAGE 3: SENTIMENT SCORING

Filtered sentences and headlines are processed through FinBERT:

- **Model:** FinBERT (Prosus AI) financial domain-specific transformer
- **Output:** Scores $\sigma_{\text{raw}} \in [-1, 1]$ for each stock are averaged over the look-back period (30-days):

d: STAGE 4: STATE INTEGRATION

Processed scores σ_t are:

- **Aligned:** Matched to the monthly rebalancing dates.
- **Embedded:** Incorporated into the state vector $\mathbf{s}_t = [\mathbf{p}_t, \sigma_t]$.

The sentiment-augmented state vector $\mathbf{s}_t = [\mathbf{p}_t, \sigma_t]$ provides the DRL framework with both traditional market signals (price data $\mathbf{p}_t$) and behavioral indicators (sentiment scores σ_t). This enhanced information set enables more informed portfolio decisions by capturing market sentiment dynamics that may precede price movements, particularly during periods of high volatility when tail risk considerations become critical for the CVaR optimization objective.

C. DEEP REINFORCEMENT LEARNING FRAMEWORK

The sentiment-augmented state vector $\mathbf{s}_t = [\mathbf{p}_t, \sigma_t]$ constructed through our pipeline serves as the foundation for our Deep Reinforcement Learning approach to CVaR diversification optimization. We employ the Deep Deterministic Policy Gradient (DDPG) algorithm with the objective of maximizing the Conditional Value-at-Risk (CVaR) diversification ratio, DR_{CVaR} . Our implementation leverages the Stable Baselines3 library, a PyTorch-based reinforcement learning framework in Python, to facilitate efficient and stable training of the DDPG algorithm. DDPG is a model-free, off-policy reinforcement learning algorithm designed for continuous action spaces, making it well-suited for portfolio allocation problems where asset weights are continuously adjustable. In this framework, the agent interacts with the financial market environment by observing historical stock prices and sentiment scores, making portfolio allocation decisions, and receiving rewards based on the optimization objective as provided in Equation (8). The learning process is guided by two neural networks: an actor network, which learns the optimal portfolio weight allocations, and a critic network, which evaluates the expected return of a given state-action pair.

1) STATE, ACTION, AND REWARD DESIGN

a: STATE SPACE

The state space at time t is defined as:

$$\mathbf{s}_t = [\mathbf{p}_t, \sigma_t], \quad (9)$$

where $\mathbf{p}_t = [p_{1,t}, p_{2,t}, \dots, p_{N,t}]$ represents normalized historical prices of N assets, and $\sigma_t = [\sigma_{1,t}, \sigma_{2,t}, \dots, \sigma_{N,t}]$ contains sentiment scores derived from a rigorous preprocessing pipeline applied to financial news.

These sentiment scores are generated by first extracting S&P 500 constituent-relevant sentences through a two-stage process using NLTK sentence tokenization and case-insensitive regex matching of company names and tickers, followed by FinBERT-based scoring to produce standardized values in $[-1, 1]$. This method preserves textual context while ensuring precise entity detection, with the resulting scores temporally aligned to market data. By incorporating these processed sentiment features into the state space, the agent gains the capacity to adapt to qualitative market signals during periods of stress, transforming sentiment from an auxiliary input into a robust, integrated component of the decision-making framework. The scores further influence the reward function, creating a closed-loop system where both quantitative and qualitative factors jointly optimize portfolio performance.

b: ACTION SPACE

The action space consists of portfolio weights assigned to each asset:

$$\mathbf{w}_t = [w_{1,t}, w_{2,t}, \dots, w_{N,t}], \quad \sum_{i=1}^N w_{i,t} = 1, \quad w_{i,t} \geq 0. \quad (10)$$

The actor network μ_θ is trained to output $\mathbf{w}_t$ given the state $\mathbf{s}_t$:

$$\mathbf{w}_t = \mu_\theta(\mathbf{s}_t). \quad (11)$$

Portfolio weights are constrained to be non-negative and sum to one, ensuring a fully invested, long-only portfolio.

c: REWARD FUNCTION

The reward $R_t : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ combines four objectives:

$$R_t = \underbrace{\text{DR}_{\text{CVaR},t}}_{\text{Diversification}} + \lambda_1 \underbrace{r_{p,t}}_{\text{Return}} + \lambda_2 \underbrace{\bar{\sigma}_t}_{\text{Sentiment}} - \lambda_3 \underbrace{C_t}_{\text{Transaction Costs}} \quad (12)$$

where:

$$\text{DR}_{\text{CVaR},t} = \frac{\sum_{i=1}^N w_{i,t} \text{CVaR}_\kappa(R_i)}{\text{CVaR}_\kappa(R_p)} \quad (\text{Eq. (7)}) \quad (13)$$

$$r_{p,t} = \mathbf{w}_t^T \mathbf{r}_t \quad (\text{Portfolio return}) \quad (14)$$

$$\bar{\sigma}_t = \mathbf{w}_t^T \sigma_t \quad (\text{Sentiment alignment}) \quad (15)$$

$$C_t = \sum_{i=1}^N c_i |w_{i,t} - w_{i,t-1}| \quad (\text{Transaction costs}) \quad (16)$$

$$\lambda_1 = 0.3, \lambda_2 = 0.1, \lambda_3 = 0.05 \quad (17)$$

- The primary term $\text{DR}_{\text{CVaR},t}$ directly optimizes the CVaR diversification ratio
- Return term encourages growth (λ_1 balances risk-return tradeoff)
- Sentiment term provides qualitative guidance (λ_2 prevents overfitting to noisy signals)
- Transaction cost penalty C_t discourages excessive rebalancing and promotes stability
- λ_i were selected via grid search over $[0, 1]$ in increments of 0.05, optimizing the Sharpe ratio.

The λ -weights ($\lambda_1 = 0.3, \lambda_2 = 0.1, \lambda_3 = 0.05$) balance the trade-offs between return maximization, sentiment alignment, and transaction cost minimization. These were selected via grid search to optimize the Sharpe ratio. Practitioners may calibrate these weights based on their risk appetite, sentiment reliability, and cost constraints. The primary term, $\text{DR}_{\text{CVaR},t}$, promotes risk diversification by maximizing the CVaR diversification ratio, thereby mitigating extreme downside risk at the confidence level κ . The return term, weighted by λ_1 , encourages capital growth while preventing excessive risk-taking. The sentiment term, weighted by $\lambda_2 \ll \lambda_1$, introduces a qualitative bias derived from news-based market sentiment, smoothed over time to reduce overfitting to noisy signals. Finally, the transaction cost penalty C_t , scaled by λ_3 , prevents the agent from over-trading by subtracting proportional costs tied to portfolio turnover, thereby improving practical deployability in real markets.

2) EXPLORATION STRATEGY

Since DDPG is an off-policy algorithm, exploration is necessary to prevent premature convergence to suboptimal strategies. We introduce an exploration noise term using the Ornstein-Uhlenbeck process:

$$\mathcal{N}_t = \theta(\mu - \mathcal{N}_{t-1}) + \sigma \epsilon_t, \quad (18)$$

where $\epsilon_t \sim \mathcal{N}(0, 1)$ is Gaussian noise. This process generates temporally correlated noise, which is beneficial for exploration in continuous action spaces.

3) DEEP DETERMINISTIC POLICY GRADIENT (DDPG)

DDPG is an actor-critic algorithm that learns deterministic policies for continuous action spaces. It maintains two networks: an actor network, which determines the optimal portfolio allocation, and a critic network, which estimates the expected return of a given action. The critic network Q_ϕ is trained using a target value computed via Temporal Difference (TD) learning:

$$y_t = R_t + \gamma Q_{\phi'}(\mathbf{s}_{t+1}, \mu_{\theta'}(\mathbf{s}_{t+1})), \quad (19)$$

where γ is the discount factor and ϕ' and θ' are target network parameters. The loss function for training the critic network is:

$$\mathcal{L}(\phi) = \mathbb{E} \left[(Q_\phi(\mathbf{s}_t, \mathbf{w}_t) - y_t)^2 \right]. \quad (20)$$

The actor network is updated using the deterministic policy gradient:

$$\nabla_{\theta} J(\theta) = \mathbb{E} [\nabla_{\mathbf{w}_t} Q_{\phi}(\mathbf{s}_t, \mathbf{w}_t) \nabla_{\theta} \mu_{\theta}(\mathbf{s}_t)]. \quad (21)$$

This update ensures that the actor selects portfolio allocations that maximize expected returns while considering risk-adjusted diversification.

4) TRAINING PROCEDURE

The training process consists of the following steps:

- 1) Initialize the actor and critic networks with random parameters.
- 2) Initialize the target networks as copies of the original networks.
- 3) For each training episode:
 - a) Observe the state $\mathbf{s}_t$, consisting of historical prices and sentiment scores.
 - b) Select the portfolio allocation $\mathbf{w}_t = \mu_{\theta}(\mathbf{s}_t) + \mathcal{N}_t$.
 - c) Execute the action and observe the next state $\mathbf{s}_{t+1}$ and reward R_t .
 - d) Store the transition $(\mathbf{s}_t, \mathbf{w}_t, R_t, \mathbf{s}_{t+1})$ in the replay buffer.
 - e) Sample a mini-batch of transitions from the replay buffer.
 - f) Compute the target value $y_t = R_t + \gamma Q_{\phi'}(\mathbf{s}_{t+1}, \mu_{\theta'}(\mathbf{s}_{t+1}))$.
 - g) Update the critic network by minimizing the loss function:

$$\mathcal{L}(\phi) = \mathbb{E} [(Q_{\phi}(\mathbf{s}_t, \mathbf{w}_t) - y_t)^2]. \quad (22)$$

- h) Update the actor network using the deterministic policy gradient:

$$\nabla_{\theta} J(\theta) = \mathbb{E} [\nabla_{\mathbf{w}_t} Q_{\phi}(\mathbf{s}_t, \mathbf{w}_t) \nabla_{\theta} \mu_{\theta}(\mathbf{s}_t)]. \quad (23)$$

- i) Update the target networks using soft updates:

$$\theta' \leftarrow \tau \theta + (1 - \tau) \theta', \quad (24)$$

$$\phi' \leftarrow \tau \phi + (1 - \tau) \phi', \quad (25)$$

where $\tau \ll 1$ controls the update rate.

- 4) Repeat the process until convergence.

The training process consists of the following steps, and its implementation is summarized in Algorithm 1 for clarity and reproducibility. The DDPG model was trained for 1×10^6 timesteps using a learning rate of 0.003. Training was conducted on a GPU-enabled Google Colab environment (NVIDIA T4, 15 GB) and completed in approximately 2.8 hours per seed (three seeds for robustness). The final actor and critic losses stabilized at approximately 4.8 and 3.6×10^{-5} , respectively, indicating convergence. The size of the replay buffer was 1×10^6 with a batch size of 100 and the evaluation occurred every 10 000 steps.

The training procedure outlined above demonstrates how the Deep Deterministic Policy Gradient (DDPG) algorithm is applied to optimize portfolio weights within the CVaR

Algorithm 1 DRL CVaR Ratio Portfolio Optimization

Input: Historical prices p_t , sentiment scores σ_t

Output: Optimal portfolio weights w^*

Initialize actor μ_{θ} and critic Q_{ϕ} ; initialize target networks and replay buffer;

for $episode = 1$ **to** E **do**

for $t = 1$ **to** T **do**

 Observe state $s_t \leftarrow [p_t, \sigma_t]$;

 Select action $w_t \leftarrow \mu_{\theta}(s_t) + \text{noise}$;

 Execute action; compute reward R_t ; observe s_{t+1} ;

 Store (s_t, w_t, R_t, s_{t+1}) in replay buffer;

 Sample mini-batch from replay buffer;

 Update critic Q_{ϕ} (TD target) and actor μ_{θ} (policy gradient);

 Soft-update target networks;

return w^*

diversification ratio framework. By iteratively updating the actor and critic networks, the agent learns to allocate assets in a way that maximizes the CVaR diversification ratio, incorporating both historical price data and sentiment scores into its decision-making process. This approach not only addresses the computational challenges of portfolio CVaR optimization but also leverages the adaptive learning capabilities of DRL to navigate complex financial environments. Building on the theoretical foundations established in this section, including the construction of the CVaR diversification ratio and its integration with the DRL framework, we now turn to the empirical performance analysis of the CVaR ratio portfolio. In the next section, we evaluate its effectiveness in comparison to other quantitative portfolios, providing insights into its real-world applicability and robustness.

IV. EMPIRICAL STUDY

A. DATA

Our empirical study uses financial text data and numerical data in the form of historical share prices.

a: NUMERICAL DATA

The numerical data, comprising historical share prices of the selected stocks, was sourced from Yahoo Finance, covering the period from January 4, 2010, to January 4, 2025. This specific time frame was intentionally selected to align with the availability of financial news and headline text data, ensuring that each portfolio rebalancing decision was informed by sentiment scores derived from the most recent financial news articles and headlines. A data-driven approach was employed to assemble a portfolio of 10 equities from the S&P 500 Index. The selection criterion was based on the frequency of stock mentions in our combined financial news and headlines data set, resulting in the top-mentioned stocks being included in the portfolio. The frequency-based

TABLE 1. Equity Returns Descriptive Statistics.

Ticker	Mean (%)	Std Dev(%)	Skewness	Kurtosis
AAPL	0.11	1.76	-0.04	5.17
MSFT	0.09	1.61	0.03	7.64
AMZN	0.12	2.06	0.26	6.24
NVDA	0.20	2.86	0.62	8.09
ORCL	0.07	1.70	0.47	15.86
GOOG	0.08	1.72	0.38	8.20
TSLA	0.22	3.62	0.37	4.88
META	0.12	2.51	0.40	21.09
JPM	0.07	1.75	0.23	9.93
BAC	0.06	2.12	0.10	9.31

stock selection approach allowed us to harness the power of sentiment analysis in portfolio construction, maximizing the impact of sentiment scores on our investment decisions. Our portfolio is composed of 10 high-profile stocks: Apple (AAPL), Microsoft (MSFT), Amazon (AMZN), NVIDIA (NVDA), Oracle (ORCL), Google (GOOG), Tesla (TSLA), Meta (META), JPMorgan Chase (JPM), and Bank of America (BAC), representing a diverse range of sectors and industries. Table 1 presents the descriptive statistics for daily returns of the 10 portfolio stocks, covering the 15-year period from January 4, 2010, to January 4, 2025.

The descriptive statistics in Table 1 highlight key differences in the return distributions of the selected stocks. Mean daily returns range from 0.06% (BAC) to 0.22% (TSLA), indicating varying levels of average daily mean returns across the stocks. TSLA and NVDA show higher returns, reflecting their growth potential, while stocks like BAC and JPM exhibit lower but more stable returns. Volatility, as measured by standard deviation, ranges from 1.61% (MSFT) to 3.62% (TSLA), with TSLA and NVDA being the most volatile, suggesting higher risk. Conversely, AAPL and MSFT display the lowest volatility, making them attractive to risk-averse investors seeking stable returns.

The skewness and kurtosis statistics provide further insights into the shape of the return distributions. Most stocks exhibit positive skewness, indicating a higher likelihood of a few extreme positive returns with frequent small losses (except for AAPL, which has slightly negative skewness). High kurtosis values for ORCL (15.86) and META (21.09) suggest their returns are prone to extreme outliers, posing additional risk for investors. In contrast, stocks like TSLA and AAPL, with relatively lower kurtosis, display more stable return distributions. These statistics underline the trade-off between risk and reward, helping investors make informed decisions based on their risk tolerance and return expectations.

b: FINANCIAL TEXT DATA

Financial text data, which is used to compute sentiment scores using the FinBERT LLM, is a combination of financial news headlines and financial news articles. While previous studies on sentiment analysis in finance have often relied on either news articles or headlines alone, our study

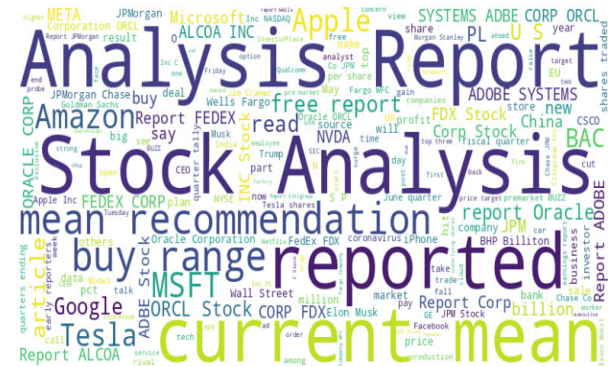


FIGURE 1. Financial News and Headlines Word Cloud.

utilizes both to capture the immediate sentiment shifts from headlines and the detailed context from articles, ensuring a more comprehensive, accurate, and timely analysis of market sentiment. Financial news headlines were obtained from daily financial news scraped of various US online sources such as the Business Daily, Seeking Alpha and Zacks Investment Research. This data is publicly available at; <https://huggingface.co/datasets/ashraq/financial-news>. Financial news articles were obtained from the FNSPID Financial News Data Set and consists of 2491 different financial news articles which covers daily market developments, analysts' stock research and companies' earnings performance results among other news. The FNSPID Financial News Data Set is publicly available at;https://github.com/Zdong104/FNSPID_Financial_News_Dataset

The word cloud in Figure 1 offers a detailed look at the combined financial text data set, highlighting the most common words and their relative frequencies, with larger words indicating more frequent occurrences. The figure demonstrates that the data set is rich in stock analysis-related content, featuring keywords such as “Stock Analysis”, “Analysis Report” and “mean recommendation”. Financial text data’s value for stock-based sentiment analysis is further enhanced by the frequent occurrence of specific stock names and tickers, including MSFT, Apple, Google, and Tesla, making it an ideal resource for this purpose.

B. FINBERT MODEL SELECTION AND ACCURACY

The combined headlines and article dataset described above was used to validate the suitability of FinBERT for our task. To benchmark performance, we evaluated sentiment classifiers on a manually labeled subset of 100 randomly sampled financial texts. As shown in Table 2, FinBERT consistently outperformed generic language models and lexicon-based approaches across all standard accuracy measures.

Accuracy captures the overall correctness of predictions, while the F1-score balances precision and recall, making it especially valuable in financial text classification where misclassifying sentiment can lead to costly portfolio errors.

TABLE 2. FinBERT vs. Alternative Sentiment Models (Test Set Performance).

Model	Accuracy	F1-Score	Precision	Recall
FinBERT	0.88	0.87	0.87	0.88
BERT-base-uncased	0.81	0.80	0.80	0.81
RoBERTa-base	0.83	0.82	0.82	0.83
VADER	0.65	0.65	0.64	0.66
TextBlob	0.63	0.62	0.61	0.63

Precision reflects the model’s ability to avoid false positives, whereas recall measures its ability to capture all relevant signals (e.g., detecting subtle bearish sentiment). FinBERT achieved an accuracy of 0.88 and an F1-score of 0.87, surpassing BERT-base and RoBERTa by a significant margin, and far exceeding traditional lexicon-based methods such as VADER and TextBlob. Its balanced precision (0.87) and recall (0.88) demonstrate both reliability and robustness in capturing nuanced sentiment from financial news. This level of precision is critical for generating dependable sentiment signals as inputs to our DRL framework. Moreover, FinBERT’s outputs (normalized to $[-1, 1]$) showed a stronger Spearman correlation with next-day returns ($\rho = 0.71$) compared to TextBlob ($\rho = 0.38$) in our preliminary analysis, underscoring the model’s economic relevance in forecasting price dynamics.

C. DRL SET-UP

Building on FinBERT’s demonstrated capability to extract economically relevant sentiment signals, we now operationalize these outputs within our DRL framework. Daily closing prices of the 10 stocks from January 4, 2010, to January 4, 2025 are used in the learning agent model training, validation and strategy backtesting. This period covers the period in which sentiment scores were available for at least one of the stocks in the portfolio. The 15 years worth of data was split using a train-validate-test ratio of 60%-20%-20% thereby ensuring that 9 years worth of data is used for training while 3 years worth of data is used for model validation and another 3 years for testing. This implies that data up to January 4, 2022 was used for training and validation (with a split between training and validation of 75%-25%) while the rest of the data was used for testing the performance of the models against the benchmarks. The train-validate-test split ratio ensures robust model training, hyperparameter tuning, and unbiased performance evaluation against benchmarks. This structured approach prevents overfitting, enhances generalizability, and simulates real-world conditions for reliable results.

The model validation stage evaluates the performance of the trained model agent with the objective of identifying and optimizing the best-performing agent. This is achieved by manually tuning the hyperparameters of the agent. For the implementation of the Deep Deterministic Policy Gradient (DDPG) model, we utilize the Stable Baselines3 reinforcement learning algorithms, which are built on the

TABLE 3. Key Parameters for DDPG in Stable Baselines3.

Parameter	Description	Value
Batch Size	Number of experiences per update.	100
Replay Buffer Size	Maximum number of stored experiences.	1, 000, 000
Tau (τ)	Soft update rate for target networks.	0.005
Exploration Noise	Gaussian noise added to actions.	$\sigma = 0.1$
Discount Factor (γ)	Importance of future rewards.	0.99
Hidden Layer Sizes	Neurons in each hidden layer.	[256, 256]
Activation Function	Non-linear function for hidden layers.	ReLU
Optimizer	Algorithm for optimizing the neural network.	Adam
Gradient Clipping	Maximum allowed gradient value.	0.5 to 5.0

PyTorch machine learning library in Python. To define the configuration of DDPG-based DRL CVaR Ratio Portfolios, the following key parameters were tuned to improve the performance of the algorithm:

- Look-back Window: The 30-day look-back window was empirically tested against the 5, 10, 60, and 90-day windows. The results showed that the 30-day window consistently provided the best balance between responsiveness and stability. This window length allowed the model to capture meaningful trends while maintaining the ability to react to recent changes, leading to improved performance metrics and more reliable predictions.
- Sentiment categorization method: Using an average of sentiment over the lookback window resulted in better outcomes compared to using the minimum or maximum sentiment. This approach provided a more stable and representative measure of sentiment, which in turn enhanced the model’s ability to make accurate predictions and informed decisions.
- Learning rate: After empirical testing, a learning rate of 0.003 was employed instead of the default value of 0.001, as it demonstrated superior performance in achieving faster convergence and better optimization results while maintaining training stability.

In addition to these hyperparameters, the DDPG algorithm utilized other parameters as detailed in Table 3 for the optimization of DRL CVaR Ratio Portfolios. These hyperparameters are standard values as specified in the Stable Baselines3 implementation documentation.

D. RESULTS AND DISCUSSION

This section investigates the performance of the our DRL CVaR Ratio Portfolios, DRL Sentiment-Aware CVaR Ratio Portfolio and the DRL Non-Sentiment-Aware CVaR Ratio Portfolio, against other portfolio benchmarks including a simple CVaR Ratio Portfolio (computed without using DRL),

TABLE 4. Portfolio Performance Comparison (Post-Transaction Costs).

Portfolio	Annualized Return (%)	Sharpe Ratio	Omega Ratio	Jensen's Alpha (%)
DRL Sentiment-Aware CVaR Ratio	34.57	1.05	1.18	0.38
DRL Non-Sentiment-Aware CVaR Ratio	28.45	0.98	1.15	0.30
Simple CVaR Ratio	24.21	0.92	1.14	0.27
Most Diversified	17.57	0.74	1.12	0.22
Min CVaR	16.78	0.55	1.08	0.28
Equally Weighted	16.18	0.46	1.01	0.17

Equally Weighted Portfolio, Most Diversified Portfolio [3] and the Minimum CVaR portfolio.

1) DRL-CVAR PORTFOLIO PERFORMANCE INSIGHTS

The performance metrics used to evaluate the portfolios include annualized return, Sharpe ratio,¹ Omega ratio,² and Jensen's alpha. Table 4 shows the great performance of the DRL CVaR Ratio portfolios with the portfolios outperforming other portfolios in terms of annualized return, Sharpe ratio, Omega ratio and Jensen's Alpha. The portfolios are rebalanced on a monthly basis to take into account changes in share prices and sentiment around the stocks.

Our DRL Sentiment-Aware CVaR Ratio Portfolio outperforms all other portfolios, achieving an annualized return of 34.57% and demonstrating superior risk-adjusted returns with a Sharpe ratio of 1.05 and an Omega ratio of 1.18. Additionally, the portfolio exhibits a higher Jensen's alpha of 0.38%. The DRL Non-Sentiment-Aware CVaR Ratio Portfolio performs well compared to other benchmark portfolios, although it does not match the performance of the sentiment-aware portfolio highlighting the importance of incorporating sentiment analysis in portfolio optimization. The superior performance of the DRL CVaR Ratio Portfolios (sentiment and non-sentiment aware) highlights the importance of using deep reinforcement learning (DRL) for portfolio optimization. DRL portfolios consistently outperform traditional portfolio strategies due to their ability to dynamically adjust asset allocations in response to changing market conditions (including share price changes and sentiment around the stock). While all portfolios are rebalanced monthly, the DRL portfolios leverage their ability to learn and adapt their strategies over time, dynamically adjusting asset allocations in response to changing market conditions (including share price changes and sentiment around the stocks). This adaptability allows DRL portfolios to achieve higher risk-adjusted returns compared to traditional portfolio strategies, which rely on static optimization methods and periodic rebalancing without the benefit of adaptive learning.

¹ Sharpe Ratio = $\frac{\mu}{\sigma}$, where μ and σ are the mean and standard deviation of the portfolio returns.

² Omega ratio = $\frac{\mu^u}{\mu^d}$, where μ^u is the absolute deviations above the benchmark and μ^d is the absolute deviations below the benchmark (We assume a performance benchmark return of 0% to calculate Omega).

The outstanding performance of our portfolios underscores the benefits of leveraging deep reinforcement learning (DRL) for portfolio optimization, as well as the advantages of maximizing the CVaR diversification ratio in portfolio construction. Portfolio optimization through maximising the diversification ratio leads to better-diversified portfolios, as the overall portfolio risk is optimized to be below the weighted average of individual asset risks. Additionally, employing CVaR instead of standard deviation in the diversification ratios ensures that the portfolio is optimized for diversification while explicitly managing downside risk. By maximizing the CVaR diversification ratio, a balance is achieved between diversification and tail-risk minimization, enhancing robustness of the portfolio in volatile market conditions. Overall, results highlight that DRL-based portfolio optimization, particularly when combined with sentiment analysis and CVaR diversification, significantly improves investment performance.

2) IMPACT OF TRANSACTION COSTS

The improved performance of the DRL CVaR ratio portfolios is achieved despite the presence of transaction costs, which negatively impact portfolio returns and are explicitly penalized in the reward function (Equation 12) to discourage excessive turnover. Figure 2 below illustrates the relationship between the drag in portfolio performance (return drag) and annual turnover percentages, highlighting a notable trend. The size of the bubbles in the graph represents the post-cost annualized return of each portfolio, scaled proportionally, where larger bubbles indicate higher annualized returns and smaller bubbles signify lower annualized returns. This visual cue allows for an immediate assessment of performance trade-offs. The DRL portfolios, such as DRL Sentiment-Aware CVaR (51.38% turnover, 0.05% drag) and DRL Non-Sentiment-Aware CVaR (46.89% turnover, 0.05% drag), exhibit relatively high turnover rates, likely due to their adaptive nature in dynamically rebalancing to optimize returns based on market conditions or sentiment data, resulting in a larger return drag. Similarly, the Equally Weighted portfolio (48.67% turnover, 0.05% drag) also demonstrates a very high turnover and drag, but its absolute performance is notably lower, as reflected by its smaller bubble size (16.18% post-cost return). In contrast, lower-turnover portfolios like Min CVaR (26.48% turnover, 0.03% drag) and Simple CVaR Ratio (30.74% turnover, 0.03% drag), depicted in brown and green, experience lower drag and deliver moderate returns (16.78% and 24.21%, respectively). The clustering of drag values between 0.03% and 0.05% indicates that the impact of transaction costs remains modest at the 10 bps level, with bubble sizes effectively underscoring the balance between turnover, drag, and overall performance.

3) RISK-ADJUSTED PERFORMANCE SENSITIVITY AND ROBUSTNESS

Complementing this analysis of transaction cost impacts, Figure 3 extends the examination to risk-adjusted performance

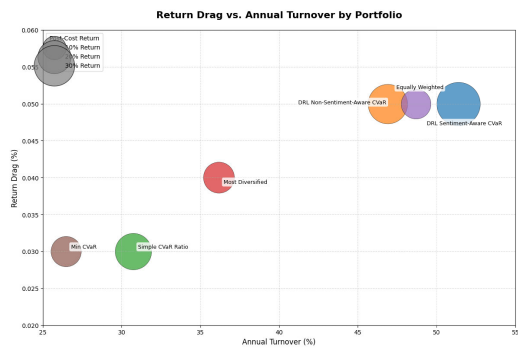


FIGURE 2. Return Drag vs. Annual Turnover by Portfolio (Assuming 10 bps per trade).

Confidence Level		Transaction Costs				
		0 bps	5 bps	10 bps	15 bps	20 bps
0,9		1,104	1,093	1,082	1,072	1,061
0,925		1,087	1,077	1,059	1,053	1,047
0,95		1,073	1,062	1,05	1,039	1,034
0,975		1,048	1,041	1,037	1,024	1,017
0,99		1,043	1,028	1,019	1,007	0,966

FIGURE 3. Sharpe Ratio Heatmap: Stress Testing CVaR Confidence (α) and Trading Costs.

by illustrating the stress testing of Sharpe ratios across different CVaR confidence levels (α) and transaction costs. This heatmap reveals how the adaptive rebalancing strategies of the DRL portfolios, while generating higher turnover, must be evaluated against the broader context of how costs erode risk-adjusted returns across various portfolio configurations. Overall, the heatmap shows two clear patterns: vertically, Sharpe ratios decline as α rises, and horizontally, Sharpe ratios erode as trading costs increase. The green shading at the top-left highlights the most favorable region, where lower confidence levels and minimal costs jointly deliver the strongest performance, providing important context for understanding the trade-offs inherent in the DRL portfolios' higher-turnover approach.

As α increases from 0.90 to 0.99, Sharpe ratios steadily decline (e.g., from 1.104 at $\alpha = 0.90$ to 1.043 at $\alpha = 0.99$ at zero cost). This pattern reflects the fact that higher α places greater weight on extreme tail losses. Since portfolio CVaR grows disproportionately due to stronger tail correlations among assets, diversification benefits weaken, causing risk-adjusted performance to deteriorate. Horizontally, Sharpe ratios also drop consistently with higher transaction costs. For example, at $\alpha = 0.95$, performance falls from 1.073 (0 bps) to 1.034 (20 bps). This erosion reflects the direct drag of trading frictions on returns, which compounds the already lower efficiency at higher α . The interaction effect between both dimensions is particularly severe at $\alpha = 0.99$: Sharpe ratios decline from 1.043 (0 bps) to just 0.966 (20 bps), effectively eliminating excess risk-adjusted gains. In contrast, moderate levels such as $\alpha = 0.925$ and $\alpha = 0.95$ appear most robust, striking a balance between tail-risk awareness and efficiency.

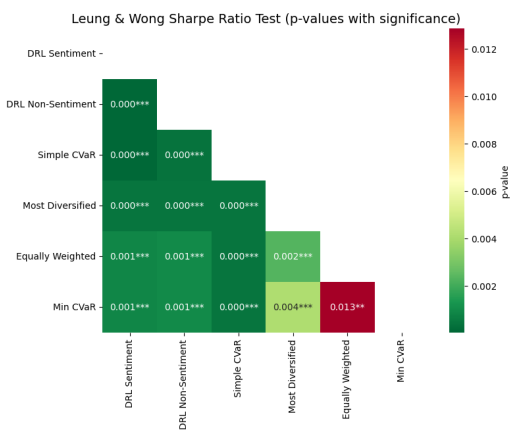


FIGURE 4. Pairwise p-values from the Leung-Wong test for differences in Sharpe ratios. Statistical significance is denoted as follows: *** p-value ≤ 0.01 , ** p-value ≤ 0.05 . The test confirms the significant outperformance of the DRL-based strategies.

While the results have shown that the DRL CVaR ratio portfolios (Sentiment-Aware and Non-Sentiment-Aware), outperform the benchmarks, it is important to establish the statistical robustness of this performance. To do that, we employ the Leung-Wong test [41], a statistical test for the significance of the difference between two Sharpe ratios. The results, summarized in Figure 4, demonstrate that the superiority of the DRL Sentiment-Aware portfolio's Sharpe ratio is highly statistically significant (p-value < 0.01) when compared to all benchmark portfolios. Furthermore, the DRL Non-Sentiment-Aware portfolio's Sharpe ratio is also significantly higher than that of the traditional strategies at the 1% significance level or better. This confirms that the outperformance is not due to chance but is a statistically reliable result of the DRL CVaR ratio methodology. The slightly higher p-value of 0.013 for the Sharpe ratio difference between the Equally Weighted and Min CVaR portfolios indicates a weaker significance level at 1% (not significant), but it remains significant at the 5% level, which is commonly accepted in finance research, suggesting a moderately distinct difference between these two strategies.

The Deep Reinforcement Learning (DRL) models employed in this study utilized the Deep Deterministic Policy Gradient (DDPG) algorithm in the Conditional Value-at-Risk (CVaR) ratio portfolio optimization process. The DDPG algorithm was selected due to its superior performance compared to other state-of-the-art DRL algorithms from the Stable Baselines 3 library, including Proximal Policy Optimization (PPO), Advantage Actor-Critic (A2C), Soft Actor-Critic (SAC), and Twin Delayed Deep Deterministic Policy Gradient (TD3). The outstanding performance of DDPG in this context is likely attributed to its ability to handle high-dimensional continuous action spaces, robustness to hyperparameter tuning, and efficient exploration-exploitation trade-off. These attributes are particularly beneficial in the dynamic and complex financial markets, where the model needs to make continuous adjustments to portfolio weights to optimize returns while managing risk effectively.

TABLE 5. Performance of DDPG against other algorithms (Sentiment-Aware) - After transaction costs.

Portfolio	Annualized Return (%)	Sharpe Ratio	Omega Ratio	Jensen's Alpha (%)
DDPG	34.57	1.05	1.18	0.38
PPO	26.65	1.04	1.16	0.30
SAC	26.26	0.95	1.14	0.30
TD3	17.58	0.67	1.08	0.24

Additionally, DDPG's deterministic policy allows for more stable and consistent portfolio allocations, which is particularly important in portfolio optimization where small changes in portfolio weights can have significant impacts on performance.

Table 5 shows the superior performance of the DDPG algorithm, which achieves a greater annualized return, better risk-adjusted returns, and a higher Jensen's Alpha. Proximal Policy Optimization (PPO) demonstrates the second-best performance in terms of all performance metrics, while the Soft Actor-Critic (SAC) comes third. Twin Delayed Deep Deterministic Policy Gradient (TD3) has the worst performance across all four algorithms, with lower risk-adjusted returns, annualized returns, and Jensen's Alpha. This poor performance of TD3 may be attributed to its increased complexity and sensitivity to hyperparameter tuning, which can lead to suboptimal results if not carefully configured. The strong performance of DDPG aligns with the conclusions of this study, as it leverages the algorithm's ability to adapt to changing market conditions, integrate sentiment analysis, and optimize for tail risk through the CVaR diversification ratio. These results further validate the proposed framework's ability to combine advanced DRL techniques with robust risk management and qualitative market insights, offering a compelling alternative to traditional portfolio optimization methods.

4) ROBUSTNESS CHECK: VALIDATION ON AN OUT-OF-SAMPLE STOCK SET

A critical test of any quantitative strategy is its performance on out-of-sample data, which helps mitigate concerns of overfitting and data snooping. To rigorously test the robustness of our DRL CVaR methodology, we validated our approach on a completely new and distinct set of assets consisting of General Electric (GE), BHP Group Limited (BHP), Netflix Inc. (NFLX), Caterpillar Inc. (CAT), Exxon Mobil Corporation (XOM), International Business Machines Corporation (IBM), General Motors Company (GM), Morgan Stanley (MS), Pfizer Inc. (PFE), and PepsiCo Inc. (PEP). This stock universe differs substantially from our in-sample portfolio, most notably in the absence of high-growth technology names such as NVIDIA (NVDA) and Tesla (TSLA), which performed exceptionally well during our backtesting period (January 2022 to January 2025), the out-of-sample set generally delivered lower annualised returns and weaker risk-adjusted performance across most metrics, reflecting the weaker market performance of these stocks

TABLE 6. Portfolio Performance with New Stock Set (Post-Transaction Costs).

Portfolio	Annualized Return (%)	Sharpe Ratio	Omega Ratio	Jensen's Alpha (%)
DRL Sentiment-Aware CVaR Ratio	22,62	0,73	1,30	0,29
DRL Non-Sentiment-Aware CVaR Ratio	21,49	0,71	1,28	0,22
Simple CVaR Ratio	16,98	0,58	1,27	0,19
Most Diversified Portfolio	12,60	0,52	1,23	0,15
Minimum CVaR	11,46	0,48	1,19	0,19
Equally Weighted	11,05	0,46	1,12	0,12

relative to the original in-sample group. The performance results for this new set are presented in Table 6.

Overall, the results confirm that our DRL-based strategies continue to outperform traditional portfolio optimization benchmarks, even when tested on a weaker stock universe. Both DRL Sentiment-Aware and Non-Sentiment-Aware CVaR Ratio portfolios exhibit superior performance compared to the classical optimization methods, though at lower levels than in the in-sample tests. This reduction in performance is expected given the absence of very strong performers such as NVDA and TSLA in the validation set.

Interestingly, while annualized returns, Sharpe ratios, and Jensen's alpha values are lower across the board, the Omega ratios for all portfolios are somewhat higher than in the in-sample tests. A probable explanation is that the out-of-sample stock set produced a return distribution with fewer extreme left-tail events (i.e., fewer large negative returns) but also lacked the very strong right-tail outcomes driven by high-growth stocks. This distributional shift results in relatively improved downside-to-upside ratios, thereby increasing the Omega ratio even though overall performance metrics are weaker. Taken together, these findings strengthen the credibility of our methodology, the DRL-based CVaR ratio portfolios maintain outperformance over standard benchmarks across different stock universes, and the shift in Omega ratio highlights how the strategy adapts to changes in the underlying return distribution. The next section of our study discusses the broader implications of these findings and outlines potential directions for future research.

V. CONCLUSION

This study introduces a groundbreaking portfolio optimization framework that synergistically combines a CVaR-based diversification ratio with Deep Reinforcement Learning (DRL) and advanced sentiment analysis based on the FinBERT LLM. By constructing a CVaR ratio portfolio that maximizes the ratio of the weighted average of individual assets' CVaRs to the overall portfolio CVaR, our approach revolutionizes traditional diversification strategies. Unlike the conventional diversification ratio method that rely solely on standard deviation as a risk metric (Most Diversified Portfolio), our framework harnesses the power of CVaR to focus on tail risks, providing a more nuanced and resilient measure of risk. This innovation addresses the inherent limitations of standard deviation in capturing extreme market

events, thereby offering a more comprehensive and robust risk assessment. Furthermore, the integration of DRL enables our framework to adapt dynamically to changing market conditions, leveraging real-time sentiment analysis to inform portfolio adjustments and enhance responsiveness.

Building on this innovative framework, our approach leverages the adaptive capabilities of the Deep Deterministic Policy Gradient (DDPG) algorithm to dynamically adjust portfolio allocations while explicitly managing tail risk through the CVaR diversification ratio. This synergy enables the model to effectively navigate complex market environments, where sentiment and risk are intricately linked. The use of DDPG allows for continuous action spaces and precise adjustments, enhancing the portfolio's ability to adapt to changing market conditions. By incorporating FinBERT-derived sentiment scores, our model captures the current market mood for portfolio assets, thereby enriching the decision-making process and fostering a more nuanced understanding of market dynamics. Notably, FinBERT-derived sentiment scores offer several advantages, including improved accuracy in capturing subtle changes in market sentiment, enhanced contextual understanding of financial text, and the ability to analyze vast amounts of unstructured data. Furthermore, FinBERT's domain-specific training data enables it to better comprehend financial jargon and nuances, leading to more informed investment decisions. This integrated approach not only enhances portfolio resilience but also provides investors with a forward-looking framework for managing risk and optimizing returns in an increasingly complex financial landscape.

Empirical analysis indicate that the proposed DRL Sentiment-Aware CVaR Ratio Portfolio outperforms traditional strategies, including the simple CVaR Ratio Portfolio, Equally Weighted Portfolios, Minimum CVaR Portfolios, and the Most Diversified Portfolio after taking transaction costs into account. The superior performance is evidenced by higher annualized returns, improved risk-adjusted returns in the form of the Sharpe and Omega ratios, and an enhanced Jensen's alpha. These results underscore the importance of using CVaR as a risk measure within the diversification ratio as it not only promotes broader risk distribution but also ensures that extreme market events are more adequately accounted for. Additionally, the results demonstrate the effectiveness of combining advanced DRL techniques with robust risk management and qualitative market insights to navigate the complexities of modern financial markets. Unlike static optimization methods, the DRL-based approach continuously learns from both the historical price data and FinBERT derived sentiment scores and adjusts its strategy accordingly, which is critical in volatile environments. The addition of sentiment analysis allows the model to capture nuances in market behavior that purely quantitative models might miss, thereby further enhancing portfolio resilience. While the proposed framework demonstrates strong performance, certain limitations must be acknowledged. First, sentiment scores derived from FinBERT may introduce biases due to

media coverage skew or entity recognition errors. Second, during crisis periods, sentiment signals can become noisy or overly reactive, potentially leading to suboptimal rebalancing decisions. Future work should explore robustness under stress scenarios and alternative sentiment sources.

For real-world implementation, monthly rebalancing is recommended to balance responsiveness with transaction cost efficiency. Training the DRL agent requires high-performance GPUs and approximately 2 to 3 hours per seed for one million timesteps; however, inference for portfolio rebalancing is lightweight and can be executed on standard servers. Despite these promising findings, several avenues for future research remain. One potential direction is to explore the integration of additional alternative data sources beyond sentiment scores derived from news and opinions about portfolio stocks. Incorporating macroeconomic indicators, geopolitical events, and other relevant data could potentially enhance the model's predictive accuracy, as these factors can also impact stock prices. Expanding the framework to include multi-asset class portfolios and accounting liquidity constraints would also improve its practical applicability. Moreover, further exploration into hyperparameter tuning and alternative DRL architectures may yield even greater improvements in performance and stability. In summary, this study demonstrates that an integrated portfolio optimization approach combining CVaR-based diversification, DRL and sentiment analysis offers a compelling and effective alternative to traditional portfolio optimization methods. The observed performance improvements suggest that such a framework not only enhances risk-adjusted returns but also provides a robust tool for adapting to the dynamic and complex nature of modern financial markets.

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